



Slurm Cheatsheet

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A compact reference for Slurm commands and useful options, with examples.

Custom CARC Slurm commands

- myaccount** - View account information for user
- noderes** - View node resources
- jobqueue** - View job queue information
- jobhist** - View compact history of user’s jobs
- jobinfo** - View detailed job information

Each command has an associated help page (e.g., **jobinfo --help**).

Job submission

- salloc** - Obtain a job allocation for interactive use ([docs](#))
- sbatch** - Submit a batch script for later execution ([docs](#))
- srun** - Obtain a job allocation and run an application ([docs](#))

Option	Description
-A, --account=<account>	Account to be charged for resources used
-a, --array=<index>	Job array specification (sbatch only)
-b, --begin=<time>	Initiate job after specified time
-C, --constraint=<features>	Required node features
--cpu-bind=<type>	Bind tasks to specific CPUs (srun only)
-c, --cpus-per-task=<count>	Number of CPUs required per task
-d, --dependency=<state:jobid>	Defer job until specified jobs reach specified state
-m, --distribution=<method[:method]>	Specify distribution methods for remote processes
-e, --error=<filename>	File in which to store job error messages (sbatch and srun only)
-x, --exclude=<name>	Specify host names to exclude from job allocation
--exclusive	Reserve all CPUs and GPUs on allocated nodes
--export=<name=value>	Export specified environment variables (e.g., all, none)
--gpus-per-task=<list>	Number of GPUs required per task
-J, --job-name=<name>	Job name
-l, --label	Prepend task ID to output (srun only)
--mail-type=<type>	E-mail notification type (e.g., begin, end, fail, requeue, all)
--mail-user=<address>	E-mail address
--mem=<size>[units]	Memory required per allocated node (e.g., 16GB)
--mem-per-cpu=<size>[units]	Memory required per allocated CPU (e.g., 2GB)
-w, --odelist=<hostnames>	Specify host names to include in job allocation
-N, --nodes=<count>	Number of nodes required for the job
-n, --ntasks=<count>	Number of tasks to be launched
--ntasks-per-node=<count>	Number of tasks to be launched per node
-o, --output=<filename>	File in which to store job output (sbatch and srun only)
-p, --partition=<names>	Partition in which to run the job
--signal=[B:]<num>[@time]	Signal job when approaching time limit
-t, --time=<time>	Limit for job run time

Examples:

```
# Request interactive job on debug node with 4 CPUs
salloc -p debug -c 4

# Request interactive job with V100 GPU
salloc -p gpu --ntasks=1 --gpus-per-task=v100:1

# Submit batch job
sbatch batch.job
```

Job management

squeue - View information about jobs in scheduling queue ([docs](#))

Option	Description
-A, --account=<account_list>	Filter by accounts (comma-separated list)
-o, --format=<options>	Output format to display
-j, --jobs=<job_id_list>	Filter by job IDs (comma-separated list)
-l, --long	Show more available information
--me	Filter by your own jobs
-n, --name=<job_name_list>	Filter by job names (comma-separated list)

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